APS106

UNIVERSITY OF TORONTO

Lists: indexing and slicing.

Week 7 | Lecture 1 (7.1)

if nothing else, write #cleancode

This Week's Content

- **Lecture 7.1**
  - **Lists: indexing and slicing**
- **Lecture 7.2**
  - Lists: nested lists and looping
- **Lecture 7.3**
  - Design Problem! Connect 4

Next Midterm: March 12

Learning objectives

- **What is a list**
- **How indexing (and slicing!) works**
- **How to modify a list**
- **Basic built-in list functions**
- **How to loop through a list**
- **How to check membership**

Motivation

We want to keep track of characters in a complex show/book

- ✓Name
- ✓Actor
- ✓Personality
- ✓Age
- ✓Title/Powers

- We could store values in a string?
- We could have unique variable names for each person?

```
gandalf_age = 24000
frodo = ”Frodo-Elijah Wood-brave, observant, and unfailingly polite-51-Ring bearer"
```

We need an efficient way to do this.

Motivation

- We could store values in a string or other individual variables?

```
gandalf_name = ”Gandalf the Grey”
gandalf_age = 24000
#all other values for Gandalf in separate variables
```

- We could have unique variable names for each person, including name, actor, personality, age and power?

```
frodo = ”Frodo-Elijah Wood-brave, observant, and unfailingly polite-51-Ring bearer”

#we could use frodo.split(‘-’) or frodo.find(‘-’) to split at or find the dashes
```

- We need a more efficient way to do this

One way: Tables or Lists!

Name	Actor	Personality	Age	Powers
Sam				
Frodo				
Gandalf				
Galadriel				
Pippin				
Aragorn				
Legolas				
Eowyn				
Gollum				
Arwen				
Merry				

Need to:

- ✓Create rows of data
- ✓Create columns of data
- ✓Be able to access a specific cell/index

Data Structures!

Data structures are "containers" that organize and group data

Built-in Python containers:

- Lists
- Sets
- Tuples
- Dictionaries

Custom containers

- Custom classes/objects

Structural/abstract containers:

- Linked lists
- Binary trees
- Stacks
- Queues
- Arrays
- Heaps

Type: List

- Can store an **ordered** collection of data using Python's type list
- The general form of a list is:

```
[val1, val2, val3, …, valN]
```

- Values are enclosed in ([ ]) and separated by commas (,)
- Can assign lists to a variable name:

```
my_list = [val1, val2, val3, …, valN]
```

List Elements

- list elements can be of any type:

```
subjects = ['bio', 'programming', 'math', 'history']
grades = [75, 98, 82, 62]
```

- A list can contain elements of more than one type:

```
street_address = [10, 'Main Street']
light = ['status',  True, 'intensity', 3.1]
```

List Operations (Indexing and Slicing)

- A list can be indexed just like a string:

```
>>> grades = [80, 90, 70, 45, 98, 57]
>>> grades[1]
90
>>> grades[-3]
45
```

- A list can be sliced just like a string:

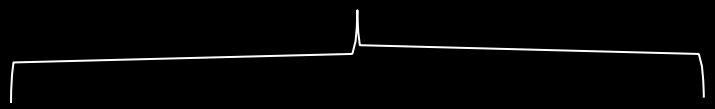
```
>>> grades[0:2]
[80, 90]
```

```
>>> grades[::-2]
[57, 45, 90]
```

Nested Lists

- Lists can contain any type, including other lists!
  - Called “nested lists”

```
[list1, list2, …, listN]
```



```
[val1, val2, …, valN]
```

- To access a nested item, first select the sublist, then treat as a regular list

```
>>> list_of_lists[0]
[val1, val2, …, valN]
>>> list_of_lists[0][1]
val2
```

Nested Lists Example

- Let's provide some information in our list of grades:

```
>>> aps106_grades = [['Midterm 1', 60],['Midterm 2', 90],['Exam', 100]]
```

```
>>> aps106_grades = [['Midterm 1', 60],
                     ['Midterm 2', 90],
                     ['Exam', 100]]
```

BOTH OF THESE ARE THE SAME THING!

- Now we can access different parts depending on what we want:

```
>>> aps106_grades[0]
['Midterm 1', 60]

>>> aps106_grades[2][1]
100
```

Nested Lists – like a table!

Name	Actor	People	Age	Powers
Sam	Sean Astin	Hobbit	33	Loyalty
Frodo	Elijah Wood	Hobbit	51	Ring bearer
Gandalf	Ian McKellen	Wizard	2400	Inspiring
Galadriel				
Pippin				
Aragorn				
Legolas				
Eowyn				
Gollum				
Arwen				
Merry				

```
[['Sam', 'Sean Astin',
'Hobbit', 33, 'Loyalty'],
['Frodo', 'Elijah Wood',
'Hobbit', 51, 'Ring
bearer'],…]
```

Same as:

```
[Sam, Frodo, …]
```

If:

```
Sam = ['Sam', 'Sean
Astin', 'Hobbit', 33,
'Loyalty']
```

Let's Code!

- Let's take a look at how this works in Python!
  - Creating lists
  - List indexing and slicing
  - List operations
  - Nested lists!

Open your notebook

Click Link:
1. The 'list' Type

List Mutability

- Lists are mutable!
  - This means they can be mutated (modified)

- All the other types we've learned so far (string, int, float, and bool) are immutable (i.e. they can NOT be modified)

```
var1 = "Mississauga"
var1[4] = "o"

??? What will happen?
```

List Mutability Example

strings are immutable

```
>>> s = "I love cats"
s[0] = "U"
Traceback (most recent call last):
builtins.TypeError: 'str' object does not
support item assignment
```

lists are mutable

```
>>> grades = [80, 90, 70, 45, 98, 57]
>>> grades[3] = 100
>>> grades[-1] = 100
>>> grades[2] = 'Perfect'
>>> grades
[80, 90, 'Perfect', 100, 98, 100]
```

Mutability is powerful! But it can get us into trouble if we're not careful.

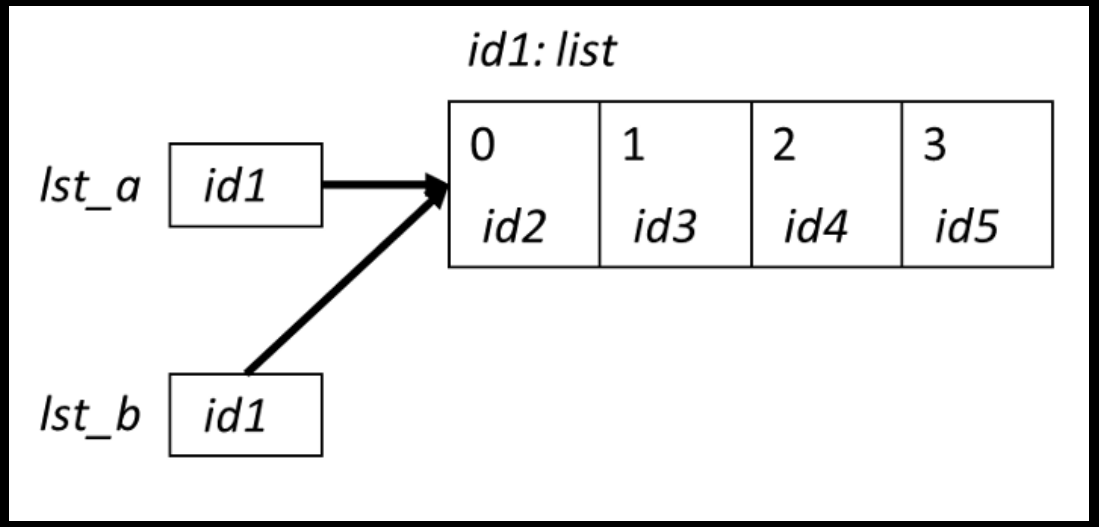
Say we want to copy a string

```
>>> lst1 =  [11, 12, 13, 14, 15, 16, 27]
>>> lst2 = lst1
>>> lst1[-1] = 17
>>> lst2
[11, 12, 13, 14, 15, 16, 17]

>>> id(lst1)
49012568
>>> id(lst2)
49012568
```

This is called aliasing

- When two variable names refer to the same object, they are aliases.
- When we modify one variable, we are modifying the object it refers to, hence also modifying the second variable.



- This is common source of error when working with list objects.

Aliasing Example (with Visualizer)

Permalink:

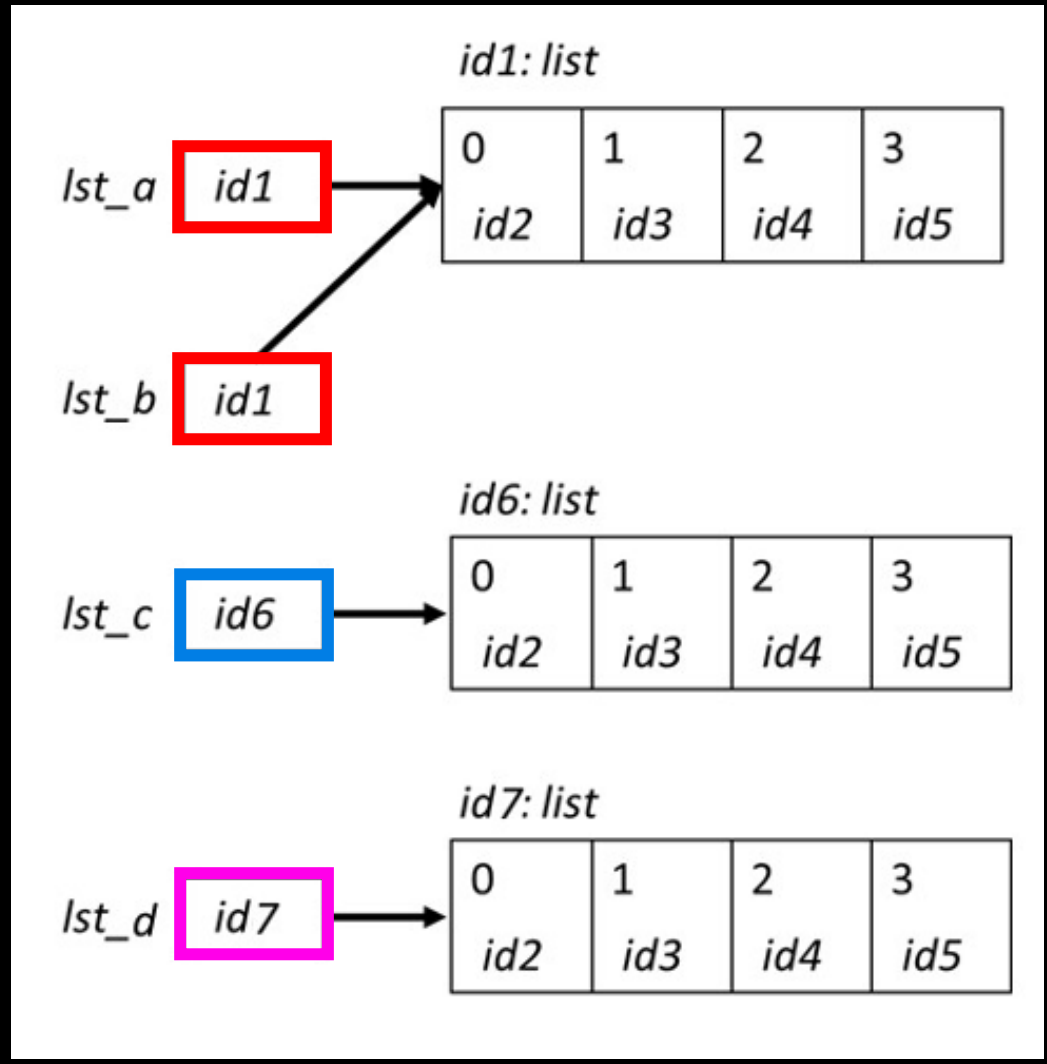
https://tinyurl.com/aps106alias

Copying Lists and Avoiding Aliasing

- There are two simple ways to copy lists:
  - Using the `list( )` function
  - Completely slice the list `[:]`

```
>>> lst_a =  [0, 1, 2, 3]
>>> lst_b = lst_a
>>> lst_c = list(lst_a)
>>> lst_d = lst_a[:]

>>> id(lst_a)
39012510
>>> id(lst_b)
39012510
>>> id(lst_c)
54514112
>>> id(lst_d)
24514139
```



Avoiding Aliasing Example (with Visualizer)

Permalink:

https://tinyurl.com/aps106alias2

Let's Code!

- Let's take a look at how this works in Python!
  - List mutability
  - Aliasing
  - Copying lists

Open your notebook

Click Link:
2. Mutability and Aliasing

Built-in Functions

- Several of Python's built-in functions can be applied to lists, including:
  - `len(list)` : return the number of elements in list (i.e. the length)
  - `min(list)` : return the value of the smallest element in list.
  - `max(list)` : return the value of the largest element in list.
  - `sum(list)` : return the sum of elements of list (list items must be numeric).

List Methods

- Lists are objects and just like other objects, the list type has associated methods that are only valid for lists
- Recall you can find out which methods are associated with objects using the built-in function dir

```
>>> dir(list)
['__add__', '__class__', '__class_getitem__', '__contains__',
'__delattr__', '__delitem__', '__dir__', '__doc__', '__eq__',
'__format__', '__ge__', '__getattribute__', '__getitem__', '__gt__',
'__hash__', '__iadd__', '__imul__', '__init__', '__init_subclass__',
'__iter__', '__le__', '__len__', '__lt__', '__mul__', '__ne__',
'__new__', '__reduce__', '__reduce_ex__', '__repr__', '__reversed__',
'__rmul__', '__setattr__', '__setitem__', '__sizeof__', '__str__',
'__subclasshook__', 'append', 'clear', 'copy', 'count', 'extend',
'index', 'insert', 'pop', 'remove', 'reverse', 'sort']
```

Adding Items to a List

- To add an object to the end of a list, use the list method append:

```
>>> colours = ['blue', 'yellow']
>>> colours.append('brown')
>>> colours
['blue', 'yellow', 'brown']
```

- To add a list to the end of a list, use the list method extend:

```
>>> colours = ['blue', 'yellow']
>>> colours.extend(['pink','green'])
>>> colours
['blue', 'yellow', 'pink', 'green']
```

Removing Items from a List

- To remove an object from a list, use the list method remove:

```
>>> colours = ['blue', 'yellow', 'pink']
>>> colours.remove('yellow')
>>> colours
['blue', 'pink']


>>> colours.remove('red')
Traceback (most recent call last):
builtins.ValueError: list.remove(x): x not in list
```

How can we write it so there's no error?

Is something in my list?

- The in operator can be used on lists too!

```
colours = ['blue', 'yellow', 'pink']

if 'red' in colours:
    colours.remove('red')
```

Let's Code!

- Let's take a look at how this works in Python!

Open your notebook

Click Link:
3. List Methods

APS106

UNIVERSITY OF TORONTO

Lists: indexing and slicing.

Week 7 | Lecture 1 (7.1)

if nothing else, write #cleancode